

Hub Operations: A Unified Framework

Concrete $\rightarrow$ Theory $\rightarrow$ Concrete

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Abstract

This document synthesizes Branch A (v1.5, operational baseline), Branch B (v2.1, idealized digital twin), and Branch C (v3.6, purposeful systems theory) into a single unified framework. It is generated using the Self-Backtracking methodology (Yang et al., 2025): the three branch documents are treated as parallel greedy decoding candidates that each reached a local optimum without fully closing the concrete-to-theory-to-concrete loop. v4.0 is the result of an explicit backtrack from all three — not an arithmetic average of their contributions, but a regeneration from the concrete operational reality upward.

The central move is methodological. The prior three branches all begin with theory and apply data. v4.0 begins with the concrete: the exact files Rafael opens during a sort, the precise parsing logic inside the HTML tools, the coordinator decision loop with its lag topology, the gap between what the tool displays and what the coordinator can act on. From this concrete reality, the framework derives theory. From theory, it returns to concrete tool specifications, data flow requirements, and a formal map of what can and cannot currently be known during a live sort.

Three new contributions distinguish v4.0 from its predecessors:

Contribution 1 — The CURE $\times$ PD Coupling (Theorem 5.1). CURE destination utilization data can be joined to the ZIP $\rightarrow$ belt truth table (the same table used by Label Training Certification) to produce per-zone dock pressure scores. When LL BEAN is at 98% utilization, this maps to a specific set of belts (Zone 9-12), which means dock-side intervention is warranted for that zone — not floor-side staffing moves. This closes the gap between Theorem 16.1 in v2.1 (the abstract bottleneck regime) and the concrete coordinator decision.

Contribution 2 — The Lag Topology (Definition 2.3). Not all data arrives at the same time. The sort has three distinct knowledge states: live (iGate only, sub-20-minute lag), same-night (SOR available mid-sort, but volume field = 0 until TMS finalizes), and next-day (SEAS, LIB, Misload, CURE all arrive t+24h). Every model variable belongs to exactly one of these three knowledge states. Mixing knowledge states produces systematically misleading numbers. v4.0 formalizes this as a partial information topology and specifies which decisions belong to which information state.

Contribution 3 — The Fidelity Cascade (Definition 6.1). Quality propagates through three measurement layers: LTC quiz accuracy $\rightarrow$ SEAS scan accuracy $\rightarrow$ LIB service failure rate. Each layer measures a different manifestation of the same upstream cause: sorters who cannot correctly route packages by ZIP produce wrong-belt scans (quiz failures), which produce misloads (SEAS events), which produce LIB events (service failures). The cascade is monotone: fidelity at the top sets an upper bound on quality at the bottom. A FidelityScore ceiling of 0.88 from LTC quiz performance bounds the minimum achievable misload rate regardless of operational pressure.

Part I — The Concrete

§1. The Coordinator’s Actual Workflow

To understand what any mathematical model must accomplish, begin with what the coordinator actually does.

The sort runs Monday through Friday, 18:00–22:00. Four hours. The coordinator for Zone 9-12 (belts PD-09, PD-10, PD-11, PD-12) arrives by 17:30 to check staffing. The laptop is open on a table at the edge of the sort floor. iGate is open in one browser tab. TMS/SOR is open in another. The hub_operations container HTML file is open in a third tab, or in a separate browser instance.

At 18:00, packages begin feeding the belts. The coordinator is on the floor.

Here is the problem that every version of this framework exists to address: **the person who should be reading and acting on the data is physically separated from the laptop the moment the data becomes meaningful.** The laptop vs. floor constraint is not a technology problem. It is a structural constraint of the operation. Coordinators manage by walking the zone, reading package flow by eye, watching faces, counting chutes.

When the coordinator does return to the laptop — typically once every 30-45 minutes — the following sequence happens:

1. **Open iGate.** Navigate to Hub Summary. Screenshot or copy the current state.
2. **Open TMS.** Navigate to SOR. Screenshot or copy the current state.
3. **Open the tracker tab.** Drag the Hub Summary and SOR files onto the upload zone. Click “Process.”
4. **Read the Hub tab.** PPH is displayed per belt. KPI tiles show hub-level throughput vs plan. Pace panel shows projected sort-down ETA.
5. **Make a decision.** If PD-11 is at 152 PPH and PD-09 is at 191 PPH, something is structurally off. Is it headcount? Chute congestion? A jam event? The coordinator walks back to the floor to investigate.
6. **Act or defer.** If headcount is the cause, the coordinator borrows from another zone. If jam events are the cause, the coordinator intervenes

physically. If the cause is unclear, the coordinator watches for one more snapshot cycle before acting.

This is the concrete coordinator decision loop. Every piece of mathematics in this framework is an attempt to make step 5 more legible — to give the coordinator a better signal about what is happening and why, so that step 6 produces better outcomes.

§2. Data Sources: What Exists, When It Arrives, and What It Actually Measures

Definition 2.1 (Data Source Taxonomy). The CHEMA operation produces data across three authority layers:

Layer 1 — Scan Authority (iGate): Every package physically scanned generates an event record in iGate. The Hub Summary and Employee Summary are real-time aggregations of these records. They are available at any point during the sort and represent the most reliable volume and productivity data available. They do not capture: hours worked, staffing assignments, quality events, or outbound dock state.

Layer 2 — Schedule Authority (SOR): The SOR (Staffing and Operations Report) is the district’s view of the operation. It contains planned hours, actual hours (when finalized), planned volume, actual volume (when finalized), area assignments, and headcount. During the sort, TMS has not finalized these values — the `actual.Volume` field is 0, and `actual.Hours` may be 0 or partial. The SOR is the authoritative source for staffing and hours; it is an unreliable source for volume. This is not a data quality failure — it is how TMS is designed.

Layer 3 — Quality Authority (SEAS, LIB, Misload, CURE): These four data sources are only available the next day. They measure what happened during the sort with a 24-hour lag. SEAS and LIB require SEAS machines to process and upload; Misload reports require overnight TMS audit processing; CURE requires trailer release and cube measurement, which occurs after the trailers leave the dock. None of these can be incorporated into a real-time decision during the sort.

Definition 2.2 (Priority Chain). When two data sources conflict on the same quantity, the tracker resolves ambiguity using a fixed priority chain:

- *Net Volume:* SOR Summary actual (if > 0) $\rightarrow \Sigma$ LooseOutbound from Hub Summary rows. When `actual.Volume` = 0 in SOR (normal during the sort), Hub Summary is the authoritative volume source.
- *Total Hours:* SOR Summary actual (if > 0) $\rightarrow \Sigma$ areaHours from SOR Staffing sheet. Hours are more reliable from SOR than from iGate because iGate tracks scans, not clock-in/clock-out.

- *PPH*: SOR Summary actual (if > 0) $\rightarrow$ Net Volume / Total Hours. Mid-sort, this is always the fallback.
- *Headcount*: SOR Summary “Worked” actual (if > 0) $\rightarrow$ distinct emp IDs from SOR staffing rows.

This priority chain exists because the two authority layers (iGate and SOR) measure different things. iGate counts what was scanned; SOR counts what was paid. They agree at sort-end (when everything has been finalized) and diverge throughout the sort. The divergence is not noise — it is information about the state of the operation.

Definition 2.3 (Lag Topology). Let $T = \{t_start, t_now, t_end, t_+24h\}$ denote the sort timeline. Define the knowledge state at time t_now as $K(t_now)$ {all data sources}:

$$\begin{aligned} K_{live}(t) &= \{\text{Hub Summary}(n), \text{Employee Summary}(n), \text{oi_archive}\} \\ K_{sornight}(t) &= K_{live}(t) \cup \{\text{SOR}(m), \text{CURE}(t - 24h)\} \\ K_{nextday} &= K_{sornight} \cup \{\text{SEAS}, \text{LIB}, \text{Misload}, \text{CURE}(t)\} \end{aligned}$$

where n is the current snapshot index (highest Hub Summary suffix so far) and m is the highest SOR suffix pulled so far.

Key property: $K_{live} \subset K_{sornight} \subset K_{nextday}$. These are nested. The coordinator during the sort operates in $K_{sornight}$. The post-sort analytics (dashboard) operate in $K_{nextday}$. Any metric that requires $K_{nextday}$ data — SQS quality components, CURE dock utilization, LIB decay — cannot inform real-time coordinator decisions. Mixing knowledge states produces systematically misleading numbers.

§3. The Tool Architecture: How Data Becomes a Number

The `hub_operations` container (v4.8.5) embeds two tools in persistent iframes: the tracker (v2.9) and the dashboard (v2.9.2). Understanding how each tool transforms raw Excel data into displayed numbers is essential for understanding what the numbers mean.

The Tracker DATA Object. All file state lives in a global DATA object:

```
DATA = {
  hub: { rows: [], excluded: [], ts, filename }, // iGate Hub Summary
  emp: { rows: [], ts, filename }, // iGate Employee Summary
  sor: { rows: [], areaHours: {}, summary: {planned:{}, actual:{}}, sortDate, ts },
  cure: { rows: [], highTfcsRows: [], gapRows: [], dateRange, ts }
}
```

Each file upload parses the Excel file, extracts rows into the corresponding DATA field, and triggers a re-render of all displayed tabs. The tracker is stateless between uploads except for IndexedDB-persisted snapshots.

****Belt Metrics Computation (`_beltMetrics`).** For every belt row from Hub Summary:

```
netVol = LooseOutbound + BagsLinked
grossVol = Gross Volume column (index 12)
pph = netVol / areaHours[belt] // from SOR; null if either is zero
pd = areaHours[belt] / distinct_emp_count_for_belt
```

This means: PPH displayed in the tracker is a function of two different data sources — iGate for the numerator (volume), SOR for the denominator (hours). If SOR hours are missing for a belt (common mid-sort for borrowed employees), PPH cannot be computed and the cell displays null. This is the most common source of coordinator confusion: the number disappears precisely when the employee situation is most complex.

Excluded Belt Logic. The following belt names are excluded from all net volume calculations:

BUILDING, AUDITS, PD13HVD, INBUILD, TOTAL (any), 999 (any)

These are meta-rows in the iGate export — administrative tracking entries that are not physical production belts. The ~5% SOR/iGate offset identified in v3.6 (Proposition 13.1) is composed precisely of these excluded types plus Air (AIRSORT) and DWSSCAN. They are not production packages. Excluding them is correct.

Coordinator Groups. The tracker aggregates belts into coordinator zones:

```
{ name: 'PD 09-12', belts: ['PD-09', 'PD-10', 'PD-11', 'PD-12'] }
```

Zone-level PPH is computed by summing all net volume across zone belts and dividing by total zone hours — not by averaging per-belt PPH. This is correct: averaging PPH would weight small belts equally with large belts. The aggregation preserves the denominator.

iGate Snapshot Architecture. The (N) suffix in Hub Summary filenames denotes the pull sequence. The tracker reads all Hub Summary files in the sort folder, orders them by (N) suffix, and uses the highest-N file as the current state. The full sequence builds the Timeline tab: each snapshot is a state vector ($V(t_n), \text{PPH}(t_n)$) at snapshot time n .

Critical rule: When the coordinator loads Hub Summary (7).xlsx, they are seeing the cumulative state as of the 7th pull, not incremental volume since pull 6. All Hub Summary files are cumulative totals, not deltas. The Timeline tab therefore displays cumulative volume vs. time, not incremental rate.

IndexedDB Persistence. The tracker stores four types of persistent records:

Store	Key	Contents
<code>hub_snapshots</code>	timestamp	Mid-sort Hub Summary snapshots
<code>area_snapshots</code>	timestamp	Mid-sort Area tab snapshots
<code>sort_history</code>	snapshotKey	End-of-sort complete records → History/Trends tabs
<code>cure_history</code>	snapshotKey	Per-sort-day CURE rows joined on sortDate

IDB Version 3. The `cure_history` store was added in v1.9 when CURE integration was first implemented. These stores persist across sessions; they survive closing and re-opening the HTML file in the same browser.

§4. The SOR and iGate as a Two-Instrument System

Proposition 4.1 (Instrument Complementarity). iGate and SOR are not redundant measures of the same quantity. They measure different aspects of the operation with different latency, authority, and resolution:

Property	iGate Hub Summary	SOR
What is measured	Packages physically scanned	Labor hours scheduled and worked
Volume authority	(primary)	(unreliable mid-sort; district allocation)
Hours authority	(scans hours)	(primary)
Real-time availability	(every pull)	Partial (actual fields = 0 mid-sort)
Area assignment	By belt	By SOR area code → belt mapping

Property	iGate Hub Summary	SOR
Borrowed employees	Appear in iGate scan location	May appear in home area in SOR

The critical insight (formalized as Proposition 13.1 in v3.6): when these two instruments disagree on volume, iGate is authoritative. When they disagree on hours, SOR is authoritative. This is not a preference — it is a consequence of the measurement design. iGate cannot know who was clocked in; SOR cannot know where packages actually went.

Corollary 4.1 (Borrowed Employee Fidelity Gap). A borrowed employee is a person who physically sorts packages on PD-09 through PD-12 (Rafael’s zone) but whose TMS clock-in is registered to another area. Their iGate scans appear in the correct location (PD-09 scan events). Their SOR hours appear in their home area. This creates a fidelity gap: iGate shows higher belt volume than the SOR hours for that belt would predict, because the denominator (SOR hours) is undercounting labor in Rafael’s zone. The PPH display for PD-09 will appear higher than the true productivity rate. The borrowed employee’s home zone will appear less productive than actual, because the numerator (iGate volume) for that zone is lower but the hours remain. Fidelity gap is structural and invisible without both sources simultaneously.

The iGate Employee Summary as Fidelity Instrument. The Employee Summary provides per-employee scan data: employee ID, belt location, packages scanned, scan PPH, hours. Comparing Employee Summary belt assignments to SOR area assignments reveals borrowed employees: employees whose SOR area code ≠ their iGate scan location. The **ghost employee** flag — an employee appearing in SOR (paid hours) with zero iGate scans — indicates an employee who did not scan (area non-participation, break without punch-out, or data entry error).

These fidelity signals are already available in the tracker; they are not yet surfaced as explicit flags. Doing so is a Tier 1 enhancement requiring no new data — only additional rendering logic in the existing DATA pipeline.

Part II — The Theory

§5. The Unified State Vector and CURE×PD Coupling

Definition 5.1 (Sort State Vector). At any snapshot time t_n during the sort, the complete operational state is given by the tuple:

$$\mathbf{s}(t_n) = (V(t_n), E(t_n), Q(t_n), L(t_n), U(t_n), \Phi(t_n))$$

where: - $V(t_n)$ = volume state: belt-level GrossVolume, netVol per belt, zone sub-totals (from iGate Hub Summary) - $E(t_n)$ = employee state: per-employee scan count, avgPPH, hours (from iGate Employee Summary) - $Q(t_n)$ = quality state: misload rates, LIB events by type, SEAS scan efficiency (from K_{nextday} only) - $L(t_n)$ = loading state: per-destination lane utilization, PPW, TFCS flags (from CURE; K_{nextday} only) - $U(t_n)$ = utilization state: planned/actual hours per area, headcount, borrow/loan counts (from SOR) - $\Phi(t_n)$ = social potential field: zone-level collective performance potential (from v3.x framework; estimated from observable proxies)

The state vector $\mathbf{s}(t_n)$ is partially observable. During the sort, the coordinator can observe $(V(t_n), E(t_n), U(t_n))$ from K_{sornight} . The quality and loading components (Q, L) are not available until K_{nextday} . The social potential $\Phi(t_n)$ is latent and estimated from the observable components via the prior weights derived in v3.4 §10.8.

Definition 5.2 (Coordinator Zone Partition). The sort belts are partitioned into coordinator zones:

$$\begin{aligned} Z_1 &= \{\text{PD-01, PD-02, PD-03, PD-04}\} \quad (\text{Gates}) \\ Z_2 &= \{\text{PD-05, PD-06, PD-07, PD-08}\} \quad (\text{Giuggio}) \\ Z_3 &= \{\text{PD-09, PD-10, PD-11, PD-12}\} \quad (\text{Almeida}) \\ Z_4 &= \{\text{SLS01, SLS02, SS}\} \quad (\text{Smalls Sort, Giatas}) \\ Z_5 &= \{\text{AIRSORT, AF02BLU, SSAIR01}\} \quad (\text{Air, Anderson}) \end{aligned}$$

The Zone 9-12 structural constant (Theorem 19.1 in v2.1) states: $V_{Z_3}(t_{\text{end}}) = 0.311 \pm 0.009 \times V_{\text{total}}$ across all sorts in the 05.04–05.08 week, regardless of nightly volume level. This is empirically verified across five sorts spanning 89,876 to 118,139 total PD packages. It is the most stable structural feature of the CHEMA sort identified to date.

Theorem 5.1 (CURE×PD Coupling — Zone Dock Pressure). *Let $D = \{d_1, \dots, d_k\}$ denote the set of active destination lanes in the CURE report. Let $\text{ZIP}^{-1}(d)$ denote the set of ZIP codes routed to destination d via the truth table, and let $\text{Belt}(\text{ZIP})$ denote the belt assignment for a given ZIP code. Define the CURE-to-belt function:*

$$B(d) = \bigcup_{\text{zip} \in \text{ZIP}^{-1}(d)} \text{Belt}(\text{zip})$$

The dock pressure for coordinator zone Z_i at time t_{+24h} is then:

$$P_{Z_i} = \frac{\sum_{d: B(d) \subseteq Z_i} U_d \cdot V_d}{\sum_{d: B(d) \subseteq Z_i} V_d}$$

where U_d is the utilization percentage for destination d and V_d is the volume loaded. When $P_{Z_i} > 0.92$ (the loading bottleneck threshold from Theorem 16.1 in v2.1), the dock-side intervention recommendation applies to zone Z_i 's coordinator: the operational bottleneck is at the trailer, not on the sort floor.

Proof sketch. The CURE file records per-destination utilization. Each destination d is a SLIC — a Sort Location Identification Code identifying a delivery center or hub. The ZIP→belt truth table in Label Training Certification maps ZIP codes to SLICs to belts. Therefore: every CURE destination row has a definite set of belts that feed it, and those belts belong to a definite coordinator zone. The dock pressure formula is a volume-weighted average of destination utilization, restricted to destinations served by a given zone's belts. When this weighted average exceeds 0.92, Theorem 16.1 applies: increasing scan rate (floor-side intervention) will not resolve the bottleneck because the dock is already saturated. The coordinator should focus on dock-side: trailer sequencing, door availability, early release of full trailers.

Corollary 5.1 (LL BEAN Structural Pressure on Zone 9-12). In the 05.04–05.08 data, LL BEAN consistently shows utilization 88–98% across the five-sort week. If LL BEAN's ZIP codes map to belts in Z_3 (which must be verified against the truth table for CHEMA's specific routing), then Zone 9-12 faces chronic dock-side pressure that manifests as PPH ceiling: sorters can clear packages quickly, but trailer loading creates a back-pressure on the dock that propagates floor-side as chute backup. This is the mechanism behind the Loading Bottleneck Regime (v2.1 §16) expressed in concrete routing terms.

Note: The exact ZIP→SLIC mapping for LL BEAN to CHEMA belts requires the SLIC→PD mapping table identified as a pending input in the ideas.md backlog. Once available, Theorem 5.1 becomes fully computable.

§6. The Fidelity Cascade

Definition 6.1 (Fidelity Cascade). The quality measurement chain across three instruments forms a monotone cascade:

Level 1 — Routing Knowledge (LTC): The Label Training Certification quiz measures whether sorters can correctly assign a ZIP code to a belt. Accuracy at Level 1 defines the routing knowledge distribution across the workforce. Define:

$$\begin{aligned} \rho_j &= \text{quiz accuracy for employee } j \\ \bar{\rho} &= \frac{1}{N} \sum_j \rho_j \quad (\text{workforce routing accuracy}) \end{aligned}$$

Level 2 — Scan Accuracy (SEAS): When a package reaches the belt, the sorter scans it. If their routing knowledge is incorrect ($\rho_j < 1$), the package may go to the wrong belt. SEAS measures the result: misloads by belt, LIB events. Define:

m_b = misload rate for belt b (from SEAS Twilight)

λ_b = LIB rate for belt b (HUB DERIVED LIB only)

Level 3 — Service Failure (SFR): When a package reaches the wrong trailer, it becomes a service failure. The Service Failure Rate (Definition 18.1 in v2.1):

$$\text{SFR} = \frac{\text{HUB DERIVED LIB}}{V_{\text{PD}}} \times 10^4 \quad (\text{per 10K packages})$$

Theorem 6.1 (Fidelity Ceiling). *For any sort with workforce routing accuracy $\bar{\rho}$, the achievable FidelityScore is bounded above:*

$$\mathcal{F} \leq \bar{\rho} \cdot (1 - \epsilon_{\text{struct}})$$

where ϵ_{struct} is a structural floor arising from physical misloads that are independent of routing knowledge (package damage, label obscurement, system outages). Empirically, $\epsilon_{\text{struct}} \approx 0.01\text{--}0.02$ at CHEMA.

Proof sketch. A sorter cannot route correctly what they do not know. If $\rho_j < 1$ for employee j , then with probability $(1 - \rho_j)$ employee j will generate a wrong-belt scan. The FidelityScore ($\mathcal{F} = \frac{1}{N} \sum_j \pi_j$ from v3.2 Theorem 4.1) is bounded by the average routing accuracy because purpose alignment (π_j) requires knowledge of correct routing. Even with full engagement and effort, an employee who does not know the ZIP→belt mapping cannot produce fidelity-aligned scans. The structural floor ϵ_{struct} captures non-knowledge causes.

Operational implication. Monitoring LIB rates and misload frequencies in the dashboard without monitoring LTC accuracy is reading Level 3 of the cascade without Level 1. When SFR is elevated, the first diagnostic question is not “was the sort rushed?” but “do the sorters on this belt know their routes?” A high quiz accuracy combined with elevated SFR is a staffing/physical signal. A low quiz accuracy combined with elevated SFR is a training signal. The cascade makes this distinction operational.

§7. Three Structural Constants

From the 05.04–05.08 empirical analysis across all three branches, three structural constants are now validated:

Constant 1: Zone 9-12 Volume Share

$$\kappa_{9-12} = 0.311 \pm 0.009$$

Observed across five consecutive sorts (range: 0.302–0.320). The stability of this constant across a 24% weekly volume decline (118,139 $\rightarrow$ 89,876 PD packages) establishes it as a routing-structural property of the CHEMA sort, not a volume-dependent quantity. This means: the coordinator for Zone 9-12 always handles approximately 31% of the night’s work, regardless of total volume. Pre-sort staffing for Zone 9-12 can be anchored to this constant: target headcount = $(V_{\text{plan}} \times 0.311) / (\text{target PPH} \times 4 \text{ hrs})$.

Constant 2: Weekly Volume Decay

$$\gamma = 0.938 \pm 0.012$$

Geometric decay in PD volume from Monday to Friday. $V(d) = V_{\text{Mon}} \times \gamma^{d-1}$ where $d = 1$ (Monday) through $d = 5$ (Friday). The Wednesday anomaly (+6.5% above -curve prediction on 05.07.26) suggests a systematic mid-week volume surge (returns processing, network rebalancing) that warrants a Wednesday-specific adjustment factor of approximately 1.07.

Constant 3: SOR/iGate Schema Offset

$$\varepsilon_{\text{schema}} = 0.050 \pm 0.010$$

The fraction of iGate-scanned packages that are non-PD (Air, DWSSCAN, AUDITS). These appear in Hub Summary rows but not in the production PD belt rows. SOR planned volume counts PD packages only. Therefore: Hub Summary GrossVolume = SOR planned volume $\times (1 + \varepsilon_{\text{schema}})$ at sort-end. This is not a discrepancy — it is a schema property. The tracker correctly excludes AUDITS and AIRSORT from net volume calculations. This constant allows cross-validation: if the observed ratio deviates from 0.05 ± 0.01 , either the sort had an unusual Air or EDS volume day, or the data has a quality issue.

§8. The Self-Backtracking Generation Method

This section documents how v4.0 was generated, as an instance of the Self-Backtracking framework applied to document synthesis.

The Self-Backtracking method (Yang et al., 2025) is originally a framework for improving language model generation quality. It formalizes the intuition that greedy generation — always choosing the locally highest-probability next token — reaches local optima that can be escaped by explicitly backtracking to an earlier state and exploring alternative continuations.

Applied to whitepaper generation across a multi-session research program:

The three branches as parallel greedy decoding. v1.5, v2.1, and v3.6 each represent a greedy continuation from a prior-version starting point. v1.5

continues v1.4 by adding empirical quality data. v2.1 continues v2.0 by adding structural constants. v3.6 continues v3.5 by applying the functor to the five-sort week. Each of these continuations is locally valid — it extends its branch consistently. But three parallel greedy decodings do not automatically converge to a unified framework.

The backtrack trigger. The trigger is the observation, after all three branches were written, that none of them fully answers the operational question: *given what I can see right now on the screen, what should I do?* v1.5 specifies what to compute. v2.1 specifies what the model predicts. v3.6 specifies what the data means in the functor framework. But the concrete interface between these — the tracker’s DATA object, the priority chain, the lag topology — appears in none of them. This is the Δ that triggers the backtrack.

The backtrack move. Instead of extending any one branch further, v4.0 backtracks to the concrete operational reality — the actual coordinator workflow, the actual tool code, the actual file naming conventions — and regenerates the framework from that ground up. This is a Type 2 backtrack in the v3.3 taxonomy: not a rejection of prior work, but a recognition that the prior candidates did not fully explore the concrete→theory→concrete direction.

The self-referential note. v4.0 is also being generated by a large language model (Claude) operating within the Cowork environment. The model’s own generation process follows the same backtracking structure: in prior sessions, the model generated v1.5 and v2.1 in parallel (two agents simultaneously), then v3.6 was completed separately. Before generating v4.0, the model ingested the KnowledgeBank.md, ideas.md, and hub_operations_v4.8.5.html architecture to ground in the concrete. The concrete-first grounding is itself a backtrack from the pattern of starting with theory (which all prior whitepapers did). The framework documents its own generation method in §8 precisely because the Self-Backtracking framework (v3.3 §10, Definition 10.8) formalized this methodology — and v4.0 is the first document generated using it explicitly.

Part III — Back to the Concrete

§9. The Sort State Machine

The sort has a definite phase structure that determines what the model can and cannot usefully compute at each moment:

Phase 0 — Pre-sort (before 18:00)

Available data: Previous sort’s CURE (from yesterday), previous sort’s SEAS/LIB (from t-24h), SOR planned values for tonight, MasterTrendAnalysis historical baselines.

Useful computations: - Pre-sort staffing estimate: $(V_{\text{plan}} \times \kappa_z) / (\text{PPH}_{\text{expected}} \times 4)$ per zone, where V_{plan} is corrected by the envelope and $\text{PPH}_{\text{expected}}$ is the day-of-

week prior from the 633-sort history - CURE dock pressure from yesterday's sort: which destinations were at capacity? This predicts tonight's loading challenge if the volume profile is similar - Zone 9-12 starting Φ_z : computed from the informed priors (v3.4 §10.8) and updated by yesterday's norm-anchor employee attendance - Pre-sort borrow/loan recommendations: which zones have excess headcount relative to volume forecast?

Phase 1 — Ramp-Up (18:00–19:00)

Available data: Hub Summary (1)–(3), Employee Summary (1)–(3), partial SOR (actual fields = 0).

Useful computations: - Early PPH trend (caution: ramp-phase PPH is structurally lower; use phase-aware thresholds) - Employee count vs. plan: who is present? - Ghost employee detection: employees in SOR with zero iGate scans after 18:20 - Zone 9-12 opening volume: is it tracking toward the $\Phi_z = 0.311$ anchor? - Φ_z update: early social dynamics signal (are norm-anchor employees present and scanning?)

Phase 2 — Steady State (19:00–21:00)

Available data: Hub Summary (4)–(16), Employee Summary (4)–(16), SOR mid-sort exports.

This is the primary diagnostic window. Full computation available for: - PPH vs. plan and vs. zone peer groups - Sort-end forecast (three scenarios: conservative 155, expected day-of-week prior, peak 188.6) - Borrow/loan suggestion: zones ahead of pace vs. zones behind, in steady-state only - Type 1 backtrack detection: $\Delta\Phi_z < _bt$ in any zone $\rightarrow$ coordinator should review prior action - Induction-driven flow entrainment check: if $_belt > \text{threshold}$ AND Φ_z flow threshold $\rightarrow$ no-action policy optimal (v3.6 §13.8)

Phase 3 — Wind-Down (21:00–22:00)

Available data: Hub Summary (17)–(21), Employee Summary (17)–(20), final SOR snapshot.

PPH rises naturally in wind-down as trailing volume flushes. Do not interpret wind-down PPH increases as performance gains. Useful computations: - Sort-down ETA: when does remaining volume reach zero at current pace? - PD-04 consolidation trigger: when should Zone 1-4 begin absorbing other zones' wrap? - WORMA N / PRORI N retain status: must be completed before sort-end - Final zone comparison for post-sort record

§10. The Coordinator Decision Loop with Model Integration

Each coordinator decision belongs to one of five action types, with a different response window and model input:

Decision Type	Response Window	Model Input	Currently Computed?
Borrow / loan	30–60 min	Zone PPH gap + pace vs plan	Manually; no auto-suggestion
Belt speed	Immediate	Jam event frequency; induction rate	Not in tracker
Area reassignment	15–30 min	Zone volume share vs Φ_z ; employee PPH	Partially (PPH display)
Break schedule	2+ hours	Fatigue curve; current PPH trajectory	Not in tracker
Sort-down call	Last 45 min	Sort-down ETA; remaining volume	Pace panel (v2.8+)

The model’s role is to reduce the time from observation to decision by making the correct action more legible from the numbers. Currently, the tracker provides the raw numbers (PPH per belt, total volume, pace). The coordinator performs the inference step manually: “PD-11 is 18% behind PD-09 at the same headcount — either there’s a jam event, or a chute backup, or the borrowed employee from yesterday didn’t show up today.” The coordinator’s brain is running the model. The goal of the tool suite is to explicitly compute and surface the model output — not to replace the coordinator’s judgment, but to ensure the judgment is applied at the right level (structural diagnosis rather than individual data point hunting).

Definition 10.1 (Model-Assisted Decision Point). A coordinator action is *model-assisted* if the model output (PPH trend, Φ_z score, sort-end forecast, CURE pressure) is explicitly consulted before the action is taken. An action is *model-independent* if the coordinator acts on direct floor observation without consulting the tracker display. Both are operationally valid. The distinction matters because model-assisted decisions are logged and can be evaluated ex post (did the action produce the predicted Δ PPH?); model-independent decisions are not automatically captured. The intervention logger (Tailscale Phase 3) makes model-independent decisions visible for the first time by logging the action and tagging the subsequent snapshot as the outcome.

§11. What the Model Cannot See

Definition 11.1 (Instrumentation Gap). An instrumentation gap is a model variable that appears in the mathematical framework but cannot be computed from any currently available data source. The following gaps are identified as of v4.0:

Model Variable	Framework Symbol	Gap Type	Closest Proxy
Belt speed	λ_{belt}	No telemetry	Induction rate from iGate timestamps
Zone norm intensity	n_z	No observation	avgPPH trajectory; norm-anchor attendance
Fatigue per employee	$F_j(t)$	No wearable	Hours-worked proxy; time-in-sort
Supervisor proximity	s_z	No location	Coordinator self-report
SLIC→PD routing	$B(d)$	Pending input	Required for Theorem 5.1
Pre-sort feeder ETA	τ_f	Telematics API needed	SOR planned arrival times
Inter-zone interaction density	ι_z	Not measured	Employee aisle assignment

The model degrades gracefully in the absence of these variables. The v3.4 §10.8 prior weights were calibrated precisely for this degradation regime — they provide the best available estimate of Φz when the direct observables are unavailable. As instrumentation gaps close (through Tailscale Phase 1-4, through SLIC→PD mapping, through belt telemetry), the model shifts from prior-dominant to data-dominant inference, following the Bayesian update protocol specified in v3.4.

§12. Synthesis: The Concrete→Theory→Concrete Loop

The concrete comes first. Not because theory is less important — Branch C (v3.x) has demonstrated that the purposeful systems theory is both mathematically precise and operationally consequential. But because theory that is not grounded in the concrete workflow becomes a parallel universe: internally consistent, externally useless.

The concrete reality of the CHEMA sort is: - A coordinator standing on the floor with no laptop access for 30-45 minute windows - Two data sources that

measure different things with different lag - Five decision types that operate on different timescales - A quality measurement system that arrives 24 hours after the decisions that could have acted on it - A workforce whose routing knowledge (LTC) sets an upstream ceiling on the quality achievable downstream (SEAS, LIB)

The theory abstracts from the concrete. The sort state vector (V, E, Q, L, U, Φ) is not arbitrary — it was built by asking: what does the coordinator need to know? Volume (V) and employee state (E) because that’s what the tracker already shows. Quality (Q) and loading (L) because they’re systematically missing from the live view. Utilization (U) because the SOR/iGate complementarity is operationally central. Social potential (Φ) because the Emergence Gap theorem proves that the interaction effects exceed individual-level variance — the zone is the unit of analysis, not the person.

The theory returns to the concrete. Theorem 5.1 (CURE×PD coupling) is not abstract: it is a computation that the dashboard’s Cube tab is one step away from performing. The Cube tab already parses CURE and computes per-destination utilization. It does not yet join that utilization to the ZIP truth table from Label Training to produce zone-level dock pressure. The missing join is the concrete implementation step that makes the abstract theorem operational.

The Fidelity Cascade (Definition 6.1, Theorem 6.1) is not abstract: it is a query across three systems that already exist (LTC quiz database, SEAS Employee data, LIB breakdown). The cascade is not yet surfaced as a single view in any tool. Creating that view is the concrete implementation step.

The weekly volume decay constant $\gamma = 0.938$ is not abstract: it is a parameter in the DOP Calculator. Currently the calculator accepts a single PPH input and a single planned volume. Adding the -corrected volume anchor (which accounts for the day-of-week effect) requires one additional input field (day index) and one parameter ($\gamma = 0.938$). The implementation is a 20-line addition to the existing calculator JavaScript.

The Zone 9-12 structural constant $\kappa_{9-12} = 0.311$ is not abstract: it is the pre-sort staffing denominator. The pre-sort staffing formula is $(V_{\text{plan}} \times 0.311) / (\text{PPH}_{\text{target}} \times 4)$. This appears in the DOP Calculator as a zone-specific allocation — currently missing from the calculator, trivially addable.

The methodology for every future version. Start with a concrete operational problem. Identify what data exists and when it arrives (lag topology). Derive the mathematical structure that makes the data informative for the decision. Specify the concrete implementation in the tool suite. Verify against empirical data. The theory lives in the middle, between the first concrete and the last concrete. It is the instrument for moving between them, not the destination.

§13. Version Genealogy and Branch Synthesis

v4.0 formally incorporates the following contributions from its three source branches:

From v1.5 (Branch A — Operational Baseline): - SEAS integration architecture (three export types; 24h lag documented in Lag Topology Definition 2.3) - SQS calibration from five-sort empirical data (misload rates 0.051–0.072%; LIB rates 0.112–0.170%) - 7-type misload taxonomy with threshold rules (AUTO MISLOAD > 20 = pre-sort structural alert) - LIB as inverse throughput signal (Theorem 17.1): HUB DERIVED LIB is the operational metric; UNASSIGNED exclusion rule - Python 8-function ingestion spec → formalized as the Phase 1 input architecture to the Lag Topology

From v2.1 (Branch B — Digital Twin): - CURE EKF augmentation → formalized as the dock pressure component $L(t)$ in the state vector - Theorem 16.1 (loading bottleneck regime) → extended to Theorem 5.1 (CURE×PD coupling) with the SLIC bridge - Zone 9-12 structural constant κ_{9-12} → constant 1 of the three structural constants (§7) - Weekly volume decay γ → constant 2 of the three structural constants (§7) - Theorem 18.1 (LIB decay property) → confirms operational use of SFR as the post-sort quality KPI - v2.0 theorem reassessment → confirms that v1.4 (and v1.5) model errors are bounded by unobserved layers (Theorem 8.1)

From v3.6 (Branch C — Purposeful Systems): - iGate data primacy (Proposition 13.1) → formalized as Proposition 4.1 (instrument complementarity) - SOR schema offset ($\varepsilon_{\text{schema}}$) → constant 3 of the three structural constants (§7) - PPH belief state update trajectory (166→188.6→155) → informs day-of-week PPH priors in the three-scenario DOP Calculator - Norm-anchor employees (11/31 = 35% in Zone 9-12) → parameterizes the restoring force in the Φz prior update - Induction-driven flow entrainment (Definition 13.4) → formalized as the no-action policy condition in Phase 2 (§9) - Tailscale Phase 1-4 confirmed deployment → directly referenced in the implementation tier sequencing (CHEMA_Analytics_Roadmap.md)

New in v4.0 (not in any prior branch): - Lag Topology (Definition 2.3): K_{live} K_{sornight} K_{nextday} - Instrument Complementarity (Proposition 4.1): iGate SOR; they measure different things - Borrowed Employee Fidelity Gap (Corollary 4.1): structural and invisible without both instruments - CURE×PD Coupling (Theorem 5.1): zone dock pressure from CURE×ZIP truth table join - Fidelity Cascade (Definition 6.1, Theorem 6.1): LTC → SEAS → LIB as monotone quality chain - Three Structural Constants (§7): κ_{9-12} , γ , $\varepsilon_{\text{schema}}$ formally collected and named - Sort State Machine (§9): phase-aware computation schedule - Self-Backtracking Generation Method (§8): documented as methodology

§14. What v4.1 Will Add

The instrumentation gaps identified in §11 are the research targets for v4.1. In priority order:

Priority 1 — SLIC→PD Mapping. This single input closes Theorem 5.1 and makes the CURE×PD dock pressure score computable. Once the mapping is available, the dashboard Cube tab can be extended in one session to show per-zone dock pressure, making the loading bottleneck recommendation (Theorem 16.1) concrete and actionable.

Priority 2 — Belt-Level Fidelity Score. Combining LTC quiz accuracy (by employee), SEAS scan accuracy (by belt), and LIB rate (by belt), the Fidelity Cascade becomes a trackable per-sort KPI. Requires a joined view across three data systems, none of which currently talks to the others. Implementation: a new Dashboard tab or an extended Overview tab that loads all three data sources and produces the cascade view.

Priority 3 — Phase-Aware PPH Thresholds. The three sort phases (Ramp, Steady-State, Wind-Down) have different structural PPH distributions. A 152 PPH reading at 18:45 (Phase 1) is normal; the same reading at 20:30 (Phase 2) is a diagnostic signal. The tracker Hub tab should label the current phase and adjust its conditional formatting accordingly. The phase boundary times (19:00, 21:00) should be configurable for non-standard sort windows.

Priority 4 — Pre-Sort Mode in the Container. The three-mode container vision (pre-sort, in-sort, post-sort) from ideas.md Strategic Orientation 4 is now mathematically grounded. The pre-sort computation is:

$$\text{Headcount}_{Z_i} = \frac{V_{\text{plan}} \times \gamma^{d-1} \times \kappa_{Z_i}}{\text{PPH}_{\text{target}}(d, Z_i) \times 4}$$

where $\kappa_{Z_3} = 0.311$, d is the day index, and $\text{PPH}_{\text{target}}(d, Z_i)$ is the day-of-week zone PPH prior from the MasterTrendAnalysis history. This requires the SLIC→PD mapping (Priority 1) to distribute volume across zones before the sort.

Glossary of v4.0 Symbols

Symbol	Definition	Source
$\mathbf{s}(t_n)$	Sort state vector at snapshot n	§5, Def 5.1
$V(t_n)$	Volume state (iGate Hub Summary)	§5
$E(t_n)$	Employee state (iGate Employee Summary)	§5

Symbol	Definition	Source
$Q(t_n)$	Quality state (SEAS/LIB/Misload)	§5
$L(t_n)$	Loading state (CURE)	§5
$U(t_n)$	Utilization state (SOR hours/headcount)	§5
$\Phi(t_n)$	Zone Social Potential field	§5; v3.x
κ_{9-12}	Zone 9-12 structural constant = 0.311 ± 0.009	§7
γ	Weekly volume decay = 0.938 ± 0.012	§7
$\varepsilon_{\text{schema}}$	SOR/iGate schema offset = 0.050 ± 0.010	§7
$B(d)$	CURE-to-belt function for destination d	§5, Thm 5.1
P_{Z_i}	Dock pressure score for zone Z_i	§5, Thm 5.1
K_{live}	Live knowledge state: iGate + oi_archive	§2, Def 2.3
K_{sornight}	Sort-night knowledge state: + SOR + prior CURE	§2, Def 2.3
K_{nextday}	Next-day knowledge state: full quality data	§2, Def 2.3
$\bar{\rho}$	Workforce routing accuracy (LTC aggregate)	§6, Def 6.1
$\varepsilon_{\text{struct}}$	Structural misload floor $0.01-0.02$	§6, Thm 6.1
SFR	Service Failure Rate = HUB DERIVED LIB / V_PD $\times$ 10^4	v2.1 Def 18.1

Session Log

Session	Date	Version	Focus
1–14	2025–2026-05	v1.0–v3.6	See whitepapers/README.md session log
15	2026-05-12	v1.5 + v2.1	Branch A and B parallel writeup; CHEMA_Analytics_Roadmap.md

Session	Date	Version	Focus
16	2026-05-12	v4.0	Unified synthesis using Self-Backtracking methodology; CURE×PD coupling; Lag Topology; Fidelity Cascade; three structural constants; sort phase state machine

References

All references from v1.4, v2.0, v2.1, v3.0–v3.6 are inherited. The following are new or primary to v4.0:

- **Yang et al. (2025)** — Self-Backtracking for Language Models. Motivates the generative methodology of v4.0 (§8); previously cited in v3.3 §10 for the coordinator MDP framework.
- **iGate Hub Summary schema** — UPS internal system; column headers: Belt, LooseOutbound, LooseInbound, ULDsInbound, In-Building, BagsCreated, BagsLinked, PkgsInLinkedBags, MissedBags, PkgsInMissedBags, BypassBagsLinked, ZeroPieceBagsLinked, **GrossVolume** (index 12), PreloadBagsNotLinked, HoldoverBagsLinked, HoldoverBagsCreated.
- **CHEMA KnowledgeBank.md** (2026-05-11) — Internal operational reference document for the CHEMA Hub Operations Intelligence Suite; primary source for the concrete DATA object architecture, belt metrics formula, coordinator group definitions, IDB schema, and CURE Cube tab parsing logic cited in §§2–4.
- **hub_operations_v4.8.5.html** (2026) — Hub Operations Container source code; JavaScript implementation of `__beltMetrics`, priority chain, excluded belt logic, and snapshot architecture cited in §3.
- **CHEMA_Analytics_Roadmap.md** (2026-05-12) — Integration roadmap for tracker/dashboard/DOP/sort-snapshot analytics; operationalizes the mathematical findings of v1.5, v2.1, v3.6 into implementation tiers.

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